Definition: ***Thermal conductivity*** *of the material is defined as the amount of heat conducted per unit area per unit time maintained at unit temperature gradient.*

Unit: $Wm^{-1}K^{-1}$

1.7 EXPRESSION FOR THERMAL CONDUCTIVITY

Let us consider a uniform rod AB with the temperatures T_1 (Hot) at end 'A' and T_2 (cold) at end B. Heat flows from hot end 'A' to the cold end 'B'. Let us consider a cross sectional area 'C' which is at a distance equal to the mean free path (λ) of the electron between the ends A and B of the rod as shown in fig. 1.5.

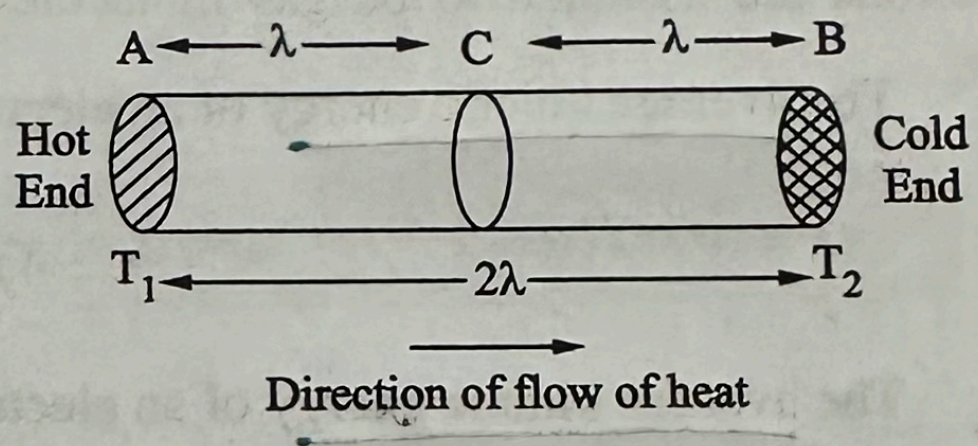

Fig. 1.5

∴ The amount of heat (Q) conducted by the rod from the end A to B of length 2λ is given by

$$Q \propto \frac{A\,(T_1 - T_2)\,t}{2\lambda}$$

(or)
$$Q = \frac{KA\,(T_1 - T_2)\,t}{2\lambda} \quad \ldots(1)$$

where $K \rightarrow$ Coefficient of thermal conductivity

$A \rightarrow$ Area of cross section of the rod

$T_1 - T_2 \rightarrow$ Temperature difference between the ends A and B

$t \rightarrow$ Time for conduction

$2\lambda \rightarrow$ Length of rod (A to B)

From equation (1) we can write

Coefficient of thermal conductivity per unit area per unit time

$$K = \frac{Q}{(T_1 - T_2)} 2\lambda \quad \ldots(2)$$

Let us assume that there is equal probability for the electrons to move in all the six directions as shown in fig. 1.6. Since each electron travels with thermal velocity v, if 'n' is the free electron density, then on an average $1/6\,nv$ electrons will travel in any one direction.

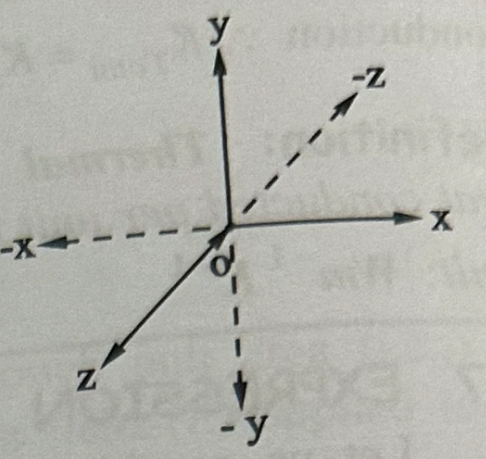

Fig. 1.6

$\therefore$ The number of electrons crossing unit area per unit time at

$$C = \frac{1}{6} nv \qquad \ldots(3)$$

According to kinetic theory of gas, [Since free electrons are assumed to be gas molecules which are freely moving]

The average kinetic energy of an electron at hot end 'A' of temperature (T_1)

$$= \frac{3}{2} K_B T_1$$

The average kinetic energy of an electron at cold end 'B' of temperature (T_2)

$$= \frac{3}{2} K_B T_2$$

where $K_B \rightarrow$ Boltzmann constant $= 1.380 \times 10^{-23}$ J/K

$\therefore$ The heat energy transferred per unit area per unit time from end A to B across C = Number of electrons × Average K.E. of electron moving from A to B

$$= \frac{1}{6} nv \cdot \frac{3}{2} K_B T_1$$

$$= \frac{1}{4} nv\, K_B T_1 \qquad \ldots(4)$$

Similarly, the heat energy transferred per unit area per unit time from end B to A across C

$$C = \frac{1}{6} nv \cdot \frac{3}{2} K_B T_2$$

$$= \frac{1}{4} nv\, K_B T_2 \qquad \ldots(5)$$

$\therefore$ The net heat energy transferred from end A to B per unit area per unit time across 'C' can be got by subtracting equation (5) from equation (4)

$$\text{(i.e.,)}\quad Q = \frac{1}{4} nvK_B T_1 - \frac{1}{4} nv\, K_B T_2$$

$$\boxed{Q = \frac{1}{4} nv\, K_B (T_1 - T_2)} \quad \text{—— (6)}$$

Substituting equation (6) in equation (2) we have

$$K = \frac{Q}{(T_1 - T_2)} \cdot 2\lambda \quad \text{—— (2)}$$

(Q value)

$$\text{Thermal conductivity } K = \frac{1}{4} \frac{nv\, K_B (T_1 - T_2) \cdot 2\lambda}{(T_1 - T_2)}$$

$$\text{(or)} \qquad K = \frac{nv\, K_B \lambda}{2} \quad \text{—— (7)}$$

We know for **metals.**

Relaxation time (τ) = Collision time (τ_c)

$$\text{(i.e.,)}\quad \tau = \tau_c = \frac{\lambda}{v}$$

$$\tau = \frac{\lambda}{v}$$

$$\text{(or)} \qquad \tau v = \lambda \quad \text{—— (8)}$$

$$\tau v = \lambda \quad \text{...(8)}$$

Substituting equation (8) in equation (7) we have

(λ value)

$$K = \frac{nv\, K_B \tau v}{2}$$

$$\text{(or) Thermal conductivity } \boxed{K = \frac{nv^2 K_B \tau}{2}} \qquad \text{...(9)}$$

Equation (9) is the classical expression for thermal conductivity.

1.8 WIEDEMANN - FRANZ LAW AND LORENTZ NUMBER

Law: *The ratio between the thermal conductivity and electrical conductivity of a metal is directly proportional to the absolute temperature of the metal.*

$$\text{(i.e.,)} \qquad \frac{K}{\sigma} \alpha\, T$$

$$\text{(or)} \qquad \boxed{\frac{K}{\sigma} = LT}$$

where L *is a constant called as* ***Lorentz number*** whose value is $2.44 \times 10^{-8}\, W\Omega K^{-2}$ (Quantum mechanical value) at temperature $T = 293$ K.

1.9 DIFFERENCES BETWEEN ELECTRICAL AND THERMAL CONDUCTIVITY

S.No.	Electrical Conductivity	Thermal Conductivity
1.	The Co-efficient of electrical conductivity is defined as the quantity of ***electricity flowing*** per unit area per unit time maintained at unit potential gradient.	The Co-efficient of Thermal conductivity is defined as the quantity of ***heat conducted*** per unit area per unit time maintained at unit temperature gradient.
2.	Electrical conductivity is purely due to number of free electrons.	Thermal conductivity is due to both free electrons and phonons.
3.	Conduction of electricity takes place from higher potential end to the lower potential end.	Conduction of heat takes place from hot end to cold end.
4.	**Unit:** $ohm^{-1} m^{-1}$	**Unit**: $Wm^{-1} K^{-1}$

1.10 SIMILARITIES BETWEEN ELECTRICAL AND THERMAL CONDUCTIVITY OF METALS

1. The electrical and thermal conductivity decreases with the increase in temperature and impurities.
2. The electrical and thermal conductivity is very high at low temperature.
3. For non-metals, the electrical and thermal conductivity is very less.

1.11 SUCCESS OF CLASSICAL FREE ELECTRON THEORY OF METALS

(i) It verifies Ohms law.

(ii) It explains the electrical and thermal conductivity of metals.

(iii) It is used to derive Wiedemann - Franz law.

(iv) The optical properties of metals can be explained using this theory.

1.12 DRAWBACKS OF CLASSICAL FREE ELECTRON THEORY

(i) It is a macroscopic theory

(ii) Classical theory states that ***all*** free electrons will absorb energy, but ***quantum theory*** states that only ***few*** electrons will absorb energy.

(iii) This theory cannot explain the Compton, photo-electric effect, paramagnetism, ferromagnetism, etc.

(iv) The theoretical and experimental values of specific heat and electronic specific heat is not matched.

(v) By classical theory $\frac{K}{\sigma T}$ = constantfor all temperatures but by Quantum theory $\frac{K}{\sigma T} \neq$ constant for all temperatures.

(vi) The Lorentz number by classical theory does not have good agreement with the experimental value and is rectified by quantum theory.